INSTITUTIONAL COURSE SYLLABUS

Course Code: Math113	Course Title: Calculus II
Course Instructor: Dr. Afifah Al-Eid	Email: aeid@psu.edu.sa
Credit Hours: 3 (3,1,0)	Course Location: On Campus
Scheduled Office Hours: S-W (10-11)	
Office Location: R226	
Co-Requisite: None	Pre-Requisite: Math111
On Campus	

Mission Statement

PSU Mission:

Prince Sultan University aims to provide the Middle East with quality education to the highest international standards. In its efforts towards successful and responsible life-long learning, PSU integrates modern technology, pedagogy, and human values for the advancement of scientific research, productivity, and leadership toward a more meaningful role in society.
PSU is committed to the effective management of institutional resources to optimize its multiple roles as a catalyst for new learning opportunities, national and international partnerships, continuous studies, professional growth, community service, and diversity in educational horizons for the good of humanity.

CHS Mission:

The College of Humanities and Sciences aspires to provide students with a high-quality education in the fields of humanities and sciences in order to advance their scientific knowledge, enhance their cultural awareness, broaden their horizons, and create a balance in student life that supports educational growth and life-long learning skills. CHS is also committed to producing quality research with a distinct societal impact.

MSD Mission:

The Department of Mathematics and Sciences is committed to offering high-quality education to lay a durable foundation for specialized degree programs at PSU. The department empowers students' academic and intellectual development by enhancing their logical thinking and analytical skills. The department adheres to the highest ethical standards and best professional practices committed to supporting research in different scientific disciplines.

I. Course Description: This course introduces various topics such as: the concept of anti-derivatives, definite and indefinite integrals, the fundamental theorem of Calculus, techniques of integration and improper integrals. Applications of definite integrals to find: area, volume, arc length and surface of revolution. Sequences and infinite series: convergence tests, integral and comparison tests, ratio and root tests, alternating series and Power series.

II. Course Learning Outcomes: On the successful completion of this course, students will be able to demonstrate the following:

Skills	**Course Learning Outcomes**
Knowledge & understanding	
Skills	**CLO1:** Evaluate the integration of basic functions such as polynomials, exponential function, logarithmic functions, etc. by using the definition of integration, anti-derivative and standard rules. ILO2, ILO5 **CLO2:** Find the area of plane region, arc length and surface of revolution by using integration techniques. ILO2, ILO5 **CLO3:** Use the integration methods to find the volume of solid generated by rotating region about vertical axis or horizontal axis. ILO2, ILO5 **CLO4:** Evaluate the integration of complicated functions by using methods of integration such as substitution, integration by parts, Trigonometric substitutions, partial fractions and trigonometric integrals. ILO2, ILO5 **CLO5:** Determine the convergence and divergence of sequences and series. ILO2, ILO5
Values	

III. Tentative Weekly Course Schedule:

Week No.	**Dates**	**UNIT/ TOPIC**	**CLO(s) alignment**	**STUDENT TASKS (assignments, exams, quizzes, presentations)**	**No. of Contact hours**
1	Dec 29 – 2 Jan	5.1 Areas 5.2 The Definite Integral	CLO1		4
2	Jan 5 - 9	5.3 The Fundamental Theorem of Calculus	CLO1		4

		5.4 Indefinite Integrals and the Net Change Theorem			
3	Jan 12 – 16	5.5 The Substitution Rule	CLO1		3
4	Jan 19 – 23	6.1 Areas Between Curves 6.2 Volumes	CLO2/ CLO3		4
5	Jan 26 – 30	6.3 Volumes by Cylindrical Shells 6.5 Average Value of a Function	CLO3		3
6	Feb 2 – 6	7.1 Integration by Parts 7.2 Trigonometric Integrals	CLO4		4
7	Feb 9 – 13	7.2 Trigonometric Integrals 7.3 Trigonometric Substitution	CLO4	Major I, Wed 12 Feb. (12-1 pm)	4
8	Feb 16 – 20	7.4 Integration of Rational Functions by Partial Fractions	CLO4		4
23 Feb Saudi Foundation Holiday					
9	Feb 24 – 27	7.5 Strategy for Integration 7.8 Improper Integrals	CLO4		4
28 Feb Beginning of Eid Alfiter Holyday					
10	Apr 6–10	8.1 Arc Length 8.2 Area of a Surface of Revolution	CLO2		4
11	Apr 13 – 17	11.1 Sequences 11.2 Series	CLO5		
12	Apr 20 – 24	11.3 The Integral Test and Estimates of Sums	CLO5	Major II, Wed 23 Apr. (12-1 pm)	4
13	Apr 27 – May 1	11.4 The Comparison Tests 11.5 Alternating Series	CLO5		4
14	May 6 – 8	11.6 Absolute Convergence and the Ratio and Root Tests 11.7 Strategy for Testing Series	CLO5		4
15	May 11 – 15	11.8 Power Series	CLO5		4
	May 17 (Saturday)	**Final Exam**			

IV. Student Assessment & Teaching Strategies: Assessment Task

Domain	Assessment Task	Teaching Strategy
Knowledge		
Skills	**Direct Assessment Tool**	Lectures.

CLO1: Evaluate the integration of basic functions such as polynomials, exponential function, logarithmic functions, etc. by using the definition of integration, anti-derivative and standard rules. CLO2: Find the area of plane region, arc length and surface of revolution by using integration techniques. CLO3: Use the integration methods to find the volume of solid generated by rotating region about vertical axis or horizontal axis. CLO4: Evaluate the integration of complicated functions by using methods of integration such as substitution, integration by parts, Trigonometric substitutions, partial fractions and trigonometric integrals. CLO5: Determine the convergence and divergence of sequences and series.	Quiz, Exams Assignments **Indirect Assessment Tool:** Course Exit Survey	Problem Solving sessions. Hands-on Practice. Examples presented in class;
Values		

V. Course Requirements: Two major exams; 4-6 quizzes; 9-10 online assignments; 1 project, Attendance & class participation.

VI. Schedule of Assessment

Assessment	Assessment Task	Week Due	Proportion of Final Assessment
1	Major I Exam	**Week 7: (Ch 5 and Ch 6)**	%20
2	Major II Exam	**Week 12: (Ch 7 and Ch8)**	%20
3	4 Quizzes	Quiz 1: Jan 15(12-1) Quiz 2: Jan 29(12-1) Quiz 3: Apr 9 (12-1) Quiz 4: May 7(12-1)	%10
4	Online Assignments & written Home work	To be announced	5 %
5	Participation and 2 Projects	To be announced	5%
6	Final Exam		%40
	Total		%100

VII. Learning Resources

A. References - required text, essential references, recommended books, electronic materials, web sites, and other learning materials. Please indicate proper bibliographic entry).

Required Textbooks:

Calculus: Early Transcendental Functions. International Metric Edition 9th edition By Stewart

Essential References Materials:

1. Calculus for Business, Economics, life Sciences, and the Social Sciences
   By: Raymond Barnett, Michael Ziegler and Karl Byleen; 11th Edition.
1. Brief Calculus for the Business, Social & life Sciences
   By: Bill Armstrong and Don Davis; 3rd Edition.

Electronic Materials:

Access to university LMS. We use the LMS to communicate with the students, Announcements and Grading are all conveyed to the students through this system.
WebAssign: www.webassign.net. We use the WEBASSIGN to create an online assignment and have a library of videos published by the author

Other Learning Materials:

Slides and/or Videos are recorded and published via LMS or links are shared on Moodle

B. Facilities Required:
laptop with camera and mic; internet

C. EDU-HUB Platform – The platform will be used for all online education experiences at PSU. It supports a variety of methods to support the teaching and learning strategies across all disciplines. It also includes the LMS.

VIII. Classroom Policies

A. Academic Integrity Policy (e.g., plagiarism or dishonesty)

"Plagiarism can be defined as unintentionally or deliberately using another person's writing or ideas as though they are one's own. Plagiarism includes, but is not limited to, copying another individual's work and taking credit for it, paraphrasing information from a source without proper documentation, and mixing one's own words with those of another author

without attribution. In addition, buying a paper or project, or downloading a paper from the Internet, and submitting them as your own is also plagiarism. The penalty for academic dishonesty will bring course expulsion and failure, or even suspension" (Academic Integrity and Syllabus Acknowledgement Form).

All students are expected to submit their 'own' work and not the work of others.

B. Attendance Policy

Please, adhere to the following guidelines:

1. The University attendance policy will be strictly followed. In this course, the absence of **(16)** hours result in a Denied Notice (DN). (Refer to the student regulation handbook for further details.)

   Please Note: IT IS THE SOLE RESPONSIBILITY OF THE STUDENT TO KEEP SATISFYING their level of attendance. Otherwise, a "DN" grade will be granted automatically during any time in the semester without any notice from the e-register system.

2. No make-up exams will be allowed except for university accepted documents. (Refer to the student regulation handbook for further details.)

C. Homework Submission Policy

It is the responsibility of the student to ensure that he/she has the Access Code and completed the Homework before the stated deadlines. Please note that after the deadlines have passed, Homework, quizzes, and majors will not be opened for any reason.

D. IX. My Assumptions (Optional)

Tips on how to enhance your problem-solving abilities:

- Attend all classes and participate in the class, it is not enough to log in, if you miss any class watch the recorded lecture that exists in your drive.
- Do all the homework assignments and the personal study plan practice on time. It is impossible to understand math without solving problems.
- You are urged to practice (but not memorize) more problems than the ones assigned.
- You should always try to solve a problem on your own before reading the solution or asking for help.
- If you find it difficult to handle a certain type of problem, you should try more problems of that type.
- You are encouraged to solve some of the review problems at the end of each chapter.
- The practice you get doing Homework and reviewing the class lectures will make exam problems easier and quicker to solve.
- Many videos are provided from the instructors of the course watch them and contact your instructor via email or google meet if you have any concerns.

	Section Number	The Mandatory Suggested Problems to be Solved
1	5.1	1,3,5
2	5.2	1,3,19,21,30,31,41,44,58,59
3	5.3	9,19,41,69,79
4	5.4	10,15,18,22,24,31,32,36,41,45,52
5	5.5	15,17,25,28,36,40,64,76
6	6.1	2,4,5,11,15,17,22
7	6.2	1,3,7,8,13,15,16,18,19,22
8	6.3	3,5,8,9,14,16,21
9	6.5	4,5,6,7,9
10	7.1	6,9,11,15,16,17,20,25,68,70,72
11	7.2	3,6,8,10,13,14,17,19,22,23,36,47
12	7.3	7,10,11,14,15,16,19,20,27,39
13	7.4	4,6,11,16,17,19,21,28,30,41,51,53
14	7.5	1,3,5,7,13,15,17,19,21,31,33,53,57,68
15	7.8	6,12,15,17,25,32,37,39,42,44,45
16	8.1	5,6,7,10,13,14,18,19,23
17	8.2	4,6,7,11,12,15,16
18	11.1	9,11,29,32,36,41,49,51,52
19	11.2	3,17,20,21,30,36,40,46,47,50

20	**11.3**	**5,8,10,13,11,18,22,27**
21	**11.4**	**12,16,19,23,26,30,32**
22	**11.5**	**11,14,19,20,27,28,30,33**
23	**11.6**	**11,12,14,19,23,24,26,28**
24	**11.7**	**5,8,17,13,23,24,43,47**
25	**11.8**	**4,7,8,9,16,19,22,25,33**